LDA classification rule (Ch4)

$$p_0 = \frac{\pi_0 f_0}{\pi_0 f_0 + \pi_1 f_1} \qquad p_1 = \frac{\pi_1 f_1}{\pi_0 f_0 + \pi_1 f_1}$$

$$f_k(x) = \frac{1}{\sqrt{(2\pi)^p |\Sigma_k|}} \exp\left\{ -\frac{1}{2} (x-\mu_k)^T \Sigma_k^{-1} (x-\mu_k) \right\}$$

assign to class k for which p_k is larger.

$\Sigma_0 = \Sigma_1$ for LDA.

$$\max(p_0, p_1) \Leftrightarrow \max(\pi_0 f_0, \pi_1 f_1)$$

$$\Leftrightarrow \max(\log\pi_0 + \log f_0, \log\pi_1 + \log f_1)$$

$$\log f_k(x) = -\frac{1}{2}(x-\mu_k)^T \Sigma^{-1}(x-\mu_k) - \log\sqrt{(2\pi)^p|\Sigma|}$$

$$= -\frac{1}{2}\left(x^T\Sigma^{-1}x - 2\mu_k^T\Sigma^{-1}x + \mu_k^T\Sigma^{-1}\mu_k \right) - \log\sqrt{(2\pi)^p|\Sigma|}$$

$\doteq \delta_0$

$$\max(p_0, p_1) \Leftrightarrow \max\left\{ \left(\log\pi_0 - \tfrac{1}{2}x^T\Sigma^{-1}x + \mu_0^T\Sigma^{-1}x - \tfrac{1}{2}\mu_0^T\Sigma^{-1}\mu_0 \right), \right.$$

$$\left. \left(\log\pi_1 - \tfrac{1}{2}x^T\Sigma x + \mu_1^T\Sigma^{-1}x - \tfrac{1}{2}\mu_1^T\Sigma^{-1}\mu_1 \right) \right\}$$

$\doteq \delta_1$

$$\delta_1 - \delta_0 = \log\pi_0 - \log\pi_0 + x^T\Sigma^{-1}(\mu_1-\mu_0) - \frac{1}{2}\left[\mu_1^T\Sigma^{-1}\mu_1 - \mu_0^T\Sigma^{-1}\mu_0 \right]$$

$$= \log\frac{\pi_1}{\pi_0} + x^T\Sigma^{-1}(\mu_1-\mu_0) - \frac{1}{2}(\mu_1+\mu_0)^T\Sigma^{-1}(\mu_1-\mu_0)$$